MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 25: Properties of some conformal mappings

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Lecture outline

1. Möbius transformations
2. Group property of Möbius transformations
3. The circle-preserving property of Möbius transformations
4. Review of the exponential function
5. The mapping $w = e^z$

Group property: Identical Möbius transformations.

The fractional linear transformation or Möbius transformation

$$w = L(z) = \frac{az+b}{cz+d} \quad (ad - bc \neq 0)$$

maps the extended plane onto itself in a one-to-one fashion. Here, we assume that the quantity $ad - bc$, called the determinant of the function $L(z)$, is nonzero, since otherwise $L(z)$ is identically equal to a constant mapping the extended plane into a single point. As we already know, the mapping $w = L(z)$ is conformal at all points of the extended plane.
Let $\mathcal{M}$ denote the set of Möbius transformations. Two such transformations

$$L_1(z) = \frac{a_1 z + b_1}{c_1 z + d_1}, \quad L_2(z) = \frac{a_2 z + b_2}{c_2 z + d_2} \tag{1}$$

are regarded as *identical* if and only if $L_1(z) = L_2(z)$ for all z.

Group property: Identity theorem. Part 1

Theorem

A necessary and sufficient condition for the two Möbius transformations (1) to be identical is that

$$a_2 = \lambda a_1,\ b_2 = \lambda b_1,\ c_2 = \lambda c_1\ d_2 = \lambda d_1\ (\lambda \neq 0). \tag{2}$$

Proof.

The sufficiency of the condition is obvious. To prove the necessity, suppose $L_1(z) = L_2(z)$. Then, in particular,

$$L_1(0) = L_2(0),\ L_1(1) = L_2(1),\ L_1(\infty) = L_2(\infty),$$

which means that

$$\frac{b_1}{d_1} = \frac{b_2}{d_2} = p,\ \frac{a_1 + b_1}{c_1 + d_1} = \frac{a_2 + b_2}{c_2 + d_2},\ \frac{a_1}{c_1} = \frac{a_2}{c_2} = q. \tag{3}$$

□

Group property: Identity theorem. Part 2

Proof.

Substituting

$$b_1 = pd_1,\ b_2 = pd_2,\ a_1 = qc_1,\ a_2 = qc_2$$

in the second of the equations (3), we obtain

$$\frac{qc_1 + pd_1}{c_1 + d_1} = \frac{qc_2 + pd_2}{c_2 + d_2}$$

or

$$(c_1 d_2 - c_2 d_1)(q - p) = 0.$$

But $q \neq p$, since otherwise

$$\frac{a_1}{c_1} = \frac{b_1}{d_1},$$

i.e.,

$$a_1 d_1 - b_1 c_1 = 0,$$

□

Group property: Identity theorem. Part 3

Proof.

contrary to the hypothesis, and therefore

$$\frac{c_1}{d_1} = \frac{c_2}{d_2},$$

Together with (3), this implies (2), as required.

□

Remark: A Möbius transformation is not characterized by the value of its determinant, since the determinant is multiplied by λ^2 when the coefficients change as described by (2). It can only be asserted that the determinant remains nonzero under any substitution (2).

Group property: Identity and inverse transformations

The transformation

$$U(z) = z,$$

which obviously belongs to the set $\mathcal{M}$, is called the *unit transformation* (or *identity transformation*). By the inverse of a given transformation

$$w = \frac{az + b}{cz + d}, \tag{4}$$

we mean the transformation which assigns to each w its inverse image z under the transformation (4). Thus the transformation

$$z = \frac{dw - b}{-cw + a}$$

(whose coefficients are unique only to within a multiplicative constant, as in Theorem (1) is the inverse of the transformation (4). We denote the inverse transformation L by L^{-1}.

Group property: Product of transformations. Part 1

Given two arbitrary Möbius transformations

$$L_1(z) = \frac{a_1 z + b_1}{c_1 z + d_1}, \quad L_2(z) = \frac{a_2 z + b_2}{c_2 z + d_2},$$

we define the *product* of L_1 by L_2 as the result of first carrying out one transformation and ten carrying out the other. There are two possible products corresponding to the two orders in which the transformations can be carried out. One product, written $L_1 L_2(z)$, equals

$$\begin{aligned} L_1 L_2(z) &= \frac{a_1\left[(a_2 z + b_2)/(c_2 z + d_2)\right] + b_1}{c_1\left[(a_2 z + b_2)/(c_2 z + d_2)\right] + d_1} \\ &= \frac{a_1(a_2 z + b_2) + b_1(c_2 z + d_2)}{c_1(a_2 z + b_2) + d_1(c_2 z + d_2)} \\ &= \frac{(a_1 a_2 + b_1 c_2)z + (a_1 b_2 + b_1 d_2)}{(c_1 a_2 + d_1 c_2)z + (c_1 b_2 + d_1 d_2)}, \end{aligned} \tag{5}$$

and the other, written $L_2 L_1(z)$, is obtained from (5) by permuting the indices 1 and 2.

Group property: Product of transformations. Part 2

Moreover, since

$$\begin{aligned}(a_1a_2+b_1c_2)(c_1b_2+d_1d_2)-(a_1b_2+b_1d_2)(c_1a_2+d_1c_2)&=\\ =(a_1d_1-b_1c_1)(a_2d_2-b_2c_2)&\neq\\ &\neq 0,\end{aligned} \tag{6}$$

each of the transformations $L_1L_2(z)$ and $L_2L_1(z)$ belongs to the set $\mathcal{M}$. In general $L_1L_2(z)\neq L_2L_1(z)$, but obviously

$$LL^{-1}(z)=L^{-1}L(z)=U(z),$$

and

$$LU(z)=UL(z)=L(z)$$

for any $L(z)\in\mathcal{M}$.

Example of the product of transformations

Example

If

$$L_1(z) = \frac{z}{z+1},$$

$$L_2(z) = \frac{z+1}{z-1},$$

then

$$L_1 L_2(z) = \frac{z+1}{2z},$$

$$L_2 L_1(z) = -2z - 1.$$

Group property: Group of transformations

Multiplication of transformations, as just defined, is an associative operation, i.e.,

$$(L_1 L_2) L_3(z) = L_1(L_2 L_3)(z). \tag{7}$$

Thus the set $\mathcal{M}$ of Möbius transformations has the following properties:

1. $\mathcal{M}$ is closed under multiplication, i.e., if $L_1 \in \mathcal{M}$, $L_2 \in \mathcal{M}$, then $L_1 L_2 \in \mathcal{M}$, $L_2 L_1 \in \mathcal{M}$.
2. Multiplication is associative.
3. There is an element $U \in \mathcal{M}$ such that $LU = UL = L$ for any $L \in \mathcal{M}$.
4. For each $L \in \mathcal{M}$, there is an element $L^{-1} \in \mathcal{M}$ such that $LL^{-1} = L^{-1}L = U$.

These four properties can be summarized by saying that $\mathcal{M}$ is a group of transformations.

The circle-preserving property of Möbius transformations 1

We now prove that any Möbius transformation carries a straight line or a circle into another straight line or circle. We call this the *circle-preserving property*, since a straight line can be regarded as a limiting case of a circle (corresponding to infinite radius). The argument consists of two parts: 1) The entire linear transformation $L(z) = \alpha z + \beta$ ($\alpha \neq 0$) is obviously a circle-preserving, since the mapping $w = L(z)$ is just a shift (if $\alpha = 1$), or a shift combined with a rotation and a uniform magnification (if $\alpha \neq 1$). 2) we need to show that denominator taking as a function if also circle-preserving:

Lemma

The transformation

$$w = \Lambda(z) = \frac{1}{z} \tag{8}$$

is circle-preserving.

The circle-preserving property of Möbius transformations 2

Proof.

The equation of any straight line or circle in the z-plane can be written in the form

$$A(x^2 + y^2) + 2Bx + 2Cy + D = 0, \tag{9}$$

where we have a straight line if $A = 0$ and at least one of the numbers B, C is nonzero, and a circle if $A \neq 0$ and $B^2 + C^2 - AD > 0$. Since

$$x^2 + y^2 = z\bar{z}, \;\; 2x = z + \bar{z}, \;\; 2y = -i(z - \bar{z}),$$

where $\bar{z} = z - iy$ is the complex conjugate of $z = x + iy$, we can rewrite (9) as

$$Az\bar{z} + \bar{E}z + E\bar{z} + D = 0, \tag{10}$$

where $E = B + iC$. It is easy to see that equation (10), where A and D are real and E is complex, is the equation of a straight line if and only if $A = 0$, $E \neq 0$, and the equation of a circle if and only if $A \neq 0$, $E\bar{E} - AD > 0$.

□

The circle-preserving property of Möbius transformations 3

Proof.

We now find the image of the curve with equation (10) under the transformation (8). Replacing z by $1/w$ in (10), we obtain

$$A\frac{1}{w\bar{w}} + \bar{E}\frac{1}{w} + E\frac{1}{\bar{w}} + D = 0$$

or

$$Dw\bar{w} + Ew + \bar{E}\bar{w} + A = 0 \tag{11}$$

Equation (11) has the same form as equation (10), with D, $\bar{E}$ and A substituted for A, E and D, respectively. It follows that (11) is the equation of a straight line if $D = 0$, since then either $A = 0$ and $E \neq 0$ if (10) is the equation of the straight line, or else $A \neq 0$ and $E\bar{E} - AD = E\bar{E} > 0$ (so that $E \neq 0$ again) if (10) is the equation of a circle. Moreover, (11) is the equation of a circle if $D \neq 0$, since then either $A \neq 0$ and $E\bar{E} - AD > 0$ if (10) is the equation of a circle, or else $A = 0$ and $E \neq 0$ (so that $E\bar{E} - AD = E\bar{E} > 0$ again) if (10) is the equation of a straight line.

□

Review of the exponential function. Part 1

An entire function which is not a polynomial is called an entire transcendental function. The simplest example of such a function is the exponential function e^z or $exp\ z$, obtained by suitably extending the familiar function e^x, defined for the real variable x, to the case where x takes arbitrary complex values, i.e., is replaced by the complex variable $z = x + iy$. The real exponential $f(x) = e^x$ is the unique function with the following properties:

1. $f(x)$ is defined and single-valued for all real x, takes only real values, and in particular takes the value e when $x = 1$;
2. $f(x)$ satisfies the *addition theorem*

$$f(x_1 + x_2) = f(x_1)f(x_2)$$

3. $f(x)$ is continuous for all x.

Review of the exponential function. Part 2

The complex function $f(z) = e^z$ can be characterized in much the same way:

Theorem

There is a unique function $f(z) = e^z$ with the following properties:

1. *$f(z)$ is defined and single-valued for all (finite) complex z, takes real values when z is real, and in particular takes the value e when $z = 1$;*
2. *$f(z)$ satisfies the addition theorem*
$$f(z_1 + z_2) = f(z_1)f(z_2) \tag{12}$$
3. *$f(z)$ is differentiable for all z, i.e. $f(z)$ is an entire function (a complex-valued function that is holomorphic over the whole complex plane).*

Proof of the exponential function theorem. Part 1

Since

$$f(z)f(1-z) = f(1) = e \neq 0$$

the function $f(z)$ cannot vanish for any z. Therefore $|f(z)| > 0$ and $Arg\, f(z)$ are defined for all z (the latter function is defined only to within an arbitrary integral multiple of 2π). It follows from (12) that

$$|f(z_1 + z_2)| = |f(z_1)|\,|f(z_2)|$$

and

$$Arg\, f(z_1 + z_2) = Arg\, f(z_1) + Arg\, f(z_2),$$

and hence the real functions of a complex variable $ln\,|f(z)|$ and $Arg\, f(z)$ both satisfy the functional equation

$$F(z_1 + z_2) = F(z_1) + F(z_2) \tag{13}$$

Proof of the exponential function theorem. Part 2

When $F(z)$ is the multiple-valued function $Arg\, f(z)$, equation (13) means that given any value of $F(z_1 + z_2)$, values of $F(z_1)$ and $F(z_2)$ can be found whose sum is $F(z_1 + z_2)$, and conversely, given any values of $F(z_1)$ and $F(z_2)$, their sum is one of the values of $F(z_1 + z_2)$.
It follows from property 3 that $f(z)$ is continuous, and therefore $ln\,|f(z)|$ and $Arg\, f(z)$ are also continuous. As applied to $Arg\, f(z)$, this means that given any values and any sequence $\{z_n\}$ converging to z, there is a sequence of values of $Arg\, f(z_n)$ such that $Arg\, f(z_n) \to Arg\, f(z)$ as $n \to \infty$. Thus $ln\,|f(z)|$ and $Arg\, f(z)$ are to be found among the continuous real solutions of (13) for the case where the independent variable is restricted to real values. In fact,

$$F(x + iy) = F(x) + F(iy) \tag{14}$$

where $F(x)$ and $F(iy)$ are functions of the real variables x and y, respectively, satisfying the equations

$$\begin{aligned} F(x_1 + x_2) &= F(x_1) + F(x_2), \\ F[i(y_1 + y_2)] &= F(iy_1) + F(iy_2). \end{aligned}$$

Proof of the exponential function theorem. Part 3

Therefore, once we know $F(x)$ and $F(iy)$ separately, we can immediately write down $F(x+iy)$ by forming the sum (14).
First suppose $F(x)$ is a single-valued continuous real function, and let $F(1) = a$. Then

$$\underbrace{F\left(\frac{1}{n}\right) + \cdots + F\left(\frac{1}{n}\right)}_{n \text{ times}} = F(1) = a,$$

which implies

$$F\left(\frac{1}{n}\right) = \frac{a}{n}$$

and

$$\underbrace{F\left(\frac{1}{n}\right) + \cdots + F\left(\frac{1}{n}\right)}_{m \text{ times}} = F\left(\frac{m}{n}\right) = \frac{m}{n}a.$$

Proof of the exponential function theorem. Part 4

Moreover

$$F(0) = 0,$$

since

$$F(x+0) = F(x) + F(0) = F(x).$$

It follows that

$$F(x) + F(-x) = F(0) = 0,$$

and hence

$$F\left(-\frac{m}{n}\right) = -\frac{m}{n}a.$$

Therefore $F(x) = ax$ for all rational x. If $X = \xi$ is irrational, and if $\{x_n\}$ is a sequence of rational numbers converging to ξ, we find that

$$F(|xi) = \lim_{n\to\infty} F(x_n) = a\xi,$$

since $F(x)$ is continuous. Thus we see that $F(x) = ax$ for all real x.

Proof of the exponential function theorem. Part 5

Similarly, we have $F(iy) = by$, so that

$$F(z) = F(x) + F(iy) = ax + by.$$

Moreover, since F is real, so are the constants a and b.
Next suppose $F(x)$ is a multiple-valued continuous real function, all of whose values are obtained from one value by adding arbitrary integral multiples of 2π, where $F(x)$ is continuous in the sense just described in connection with the function $Arg\, f(z)$. Then from the relation

$$F(x) + F(0) = F(x),$$

we deduce that the values of $F(0)$ are all integral multiples of 2π, and hence that there exists a value of $F(0)$ equal to 0. Therefore, since $F(x)$ is continuous, there must exist a $\delta > 0$ such that

$$|F(x)| < \frac{\pi}{2} \tag{15}$$

for all $|x| \leq \delta$. Because of the way in which $F(x)$ is multiple-valued, it follows that $F(x)$ has only one value satisfying (15). Let this value be denoted by $\Phi(x)$.

Proof of the exponential function theorem. Part 6

Obviously, $\Phi(x)$ is continuous (in the ordinary sense) in the interval $[-\delta, \delta]$. If x_1 and x_2 in $[-\delta, \delta]$ such that $x_1 + x_2$ also belong to $[-\delta, \delta]$, then $\Phi(x_1) + \Phi(x_2)$, being of the values of $F(x_1) + F(x_2) = F(x_1 + x_2)$, can differ from $\Phi(x_1 + x_2)$ only by a multiple of 2π. But

$$|\Phi(x_1) + \Phi(x_2)| < \pi, \quad |\Phi(x_1 + x_2)| < \pi,$$

and therefore

$$\Phi(x_1 + x_2) = \Phi(x_1) + \Phi(x_2). \tag{16}$$

Writing $\Phi(\delta) = \alpha\delta$ and applying our previous argument to the single-valued continuous function $\Phi(x)$ satisfying the functional equation (16), we find that

$$\Phi(x) = \alpha x$$

for $|x| \leq \delta$. If we use this formula to *define* $\Phi(x)$ for arbitrary real x, then, as before, one of the values of $F(x)$ equals $\Phi(x)$. In fact, any $x \geq 0$ can be uniquely represented in the form $2n\delta + \xi$, where n is a nonnegative integer and $|\xi| \leq \delta$.

Proof of the exponential function theorem. Part 7

Therefore, if $x \geq 0$, one value of the sum

$$\underbrace{F(\delta) + \cdots + F(\delta)}_{2n \text{ times}} + F(\xi) = F(x)$$

is $2n\Phi(\delta) + \Phi(\xi) = \Phi(x)$. In other words, if $x \geq 0$, all the values of $F(x)$ are of the form

$$\Phi(x) + 2j\pi \ \ (j = 0, \pm 1, \pm 2, \ldots).$$

On the other hand, if $x \leq 0$, we deduce from the relation

$$F(x) + F(-x) = F(0) = 2k\pi \ \ (k = 0, \pm 1, \pm 2, \ldots)$$

that

$$F(x) = 2k\pi - F(-x) = 2k\pi - \Phi(-x) - 2j\pi = \Phi(x) + 2(k-j)\pi = \Phi(x) + 2l\pi$$

$$(k, j, l = 0, \pm 1, \pm 2, \ldots).$$

Proof of the exponential function theorem. Part 8

Thus, if $F(x)$ is a multiple-valued function of the type under discussion, we have

$$F(x) = \alpha x + 2j\pi \ \ (j = 0, \pm 1, \pm 2, \ldots)$$

for all real x. Similarly, we have

$$F(iy) = \beta y + 2k\pi \ \ (k = 0, \pm 1, \pm 2, \ldots),$$

so that

$$F(z) = F(x) + F(iy) = \alpha x + \beta y + 2l\pi \ \ (l = 0, \pm 1, \pm 2, \ldots).$$

Moreover since F is real, so are the constants α and β.
It follows from these considerations that any single-valued continuous real function satisfying (13) must have the form $ax + by$, whereas any multiple-valued continuous real function satisfying (13), with the "same kind of multiple-valuedness" as $Arg\, f(z)$, must have the form $\alpha x + \beta y + 2l\pi\, (l = 0, \pm 1, \pm 2, \ldots)$, where a, b, α and β are all real constants.

Proof of the exponential function theorem. Part 9

Therefore, any continuous complex function which satisfies property 2 (the addition theorem), and is not identically zero or unity, nust be of the form

$$\begin{aligned} f(z) &= e^{\ln|f(z)|} e^{iArg\, f(z)} \\ &= e^{ax+by}[\cos(\alpha x + \beta y) + i\sin(\alpha x + \beta y)]. \end{aligned} \tag{17}$$

We now apply property 1, which states that $f(z)$ takes real values when z is real. Setting $z = x$, $y = 0$ in (17), we see that $\cos\alpha x + i\sin\alpha x$ must be real for all x, which implies $\alpha = 0$. Moreover, property 1 also requires that $f(1) = e$, so that $e^a = e$ or $a = 1$. Therefore any continuous function with properties 1 and 2 must have the form

$$f(z) = e^{x+by}(\cos\beta y + i\sin\beta y). \tag{18}$$

Proof of the exponential function theorem. Part 10

Finally we impose the full force of property 3 upon $f(z)$, i.e. we require that $f(z)$ be differentiable. Applying the Cauchy-Riemann equations to the function (19), we find that

$$\begin{aligned} e^{x+by}\cos\beta y &= be^{x+by}\sin\beta y + \beta e^{x+by}\cos\beta y, \\ be^{x+by}\cos\beta y - \beta e^{x+by}\sin\beta y &= -e^{x+by}\sin\beta y, \end{aligned}$$

or

$$b\sin\beta y + (\beta-1)\cos\beta y = 0, \tag{19}$$

$$b\cos\beta y - (\beta-1)\sin\beta y = 0. \tag{20}$$

Each of these relations by itself is sufficient to determine b and β. In fact, setting first $y=0$ and then $y=\pi/2$ in (20), we find that $\beta=1$ and $b=0$, and these values also satisfy (19 and 20).

Proof of the exponential function theorem. Part 11

Thus, finally, the unique function $f(z) = e^z$ with all the properties listed in the statement of the theorem is

$$f(z) = e^z = e^x(\cos y + i \sin y) \tag{21}$$

Moreover, the fact that (21) has all the desired properties is obvious by inspection.

Remark. If instead of requiring $f(z)$ to be differentiable (property 3), we had only required $f(z)$ to be continuous, then we would have obtained an infinite set of functions of the form (19), where b and β are arbitrary real numbers. However, only the function (21) is entire, i.e., analytic everywhere in the finite plane.

The mapping $w = e^z$. Part 1

It follows from the formula

$$e^z = e^x(\cos y + i \sin y) \tag{22}$$

that e^z is nonzero for all z and

$$|e^z| = e^x, \; Arg\, e^z = y + 2k\pi.$$

For $z = iy$ $(x = 0)$ we obtain Euler's formula

$$e^{iy} = \cos y + i \sin y. \tag{23}$$

Using (23), we can replace the trigonometric form of a complex number by the more concise *polar form*

$$z = re^{i\Phi}$$

The mapping $w = e^z$. Part 2

It is apparent from (21) that the exponential is periodic in z with period $2\pi i$. In other words, if z is changed by $2\pi i$, so that y is changed by 2π, the value of e^z does not change:

$$e^{z+2\pi i} = e^z$$

We now show that $2\pi i$ is the fundamental (or primitive) period of the function e^z, i.e., that any other period ω of e^z must be of the form $2k\pi i$, where k is an integer. To see this, let $\omega = \alpha + i\beta$. Then

$$e^{z+\omega} = e^z$$

for any z, and in particular,

$$e^{\omega} = e^{\alpha+i\beta} = e^{\alpha}(\cos\beta + i\sin\beta) = 1$$

for $z = 0$.

The mapping $w = e^z$. Part 3

But this means that $|e^\omega| = e^\alpha = 1$ which implies $\alpha = 0$, and hence $\cos\beta + i\sin\beta = 1$ which implies $\beta = 2k\pi i$, so that

$$\omega = \alpha + i\beta = 2k\pi i,$$

is asserted.
The expression e^∞ will be regarded as meaningless, since

$$\lim_{z\to\infty} e^z$$

does not exist. This can be seen from the fact that $e^x \to \infty$ as $x \to +\infty$, whereas $e^x \to 0$ as $x \to -\infty$. in particular, it follows that e^x cannot coincide with any polynomial, i.e., e^z is actually an entire *transcendental* function, since any polynomial (excluding the trivial case of a constant) approaches infinity as $z \to \infty$.

The mapping $w = e^z$. Part 4

Next we study the geometric behavior of the mapping $w = e^z$. e^z is nonzero for all z. This means that the origin of coordinates in the w-plane does not belong to the image of the finite z-plane under the mapping $w = e^z$. However, as we now show, any other finite point of the w-plane does belong to this image. In fact, from the equation $w = e^z$, where $w \neq 0$ is given and $z = x + iy$ is unknown, we obtain

$$|w| = e^x \text{ or } x = \ln|x|$$

and

$$Arg\, w = y + 2k\pi \text{ or } y = Arg\, w.$$

Therefore the inverse images of the point w can only be points of the form

$$z = \ln|w| + i\, Arg\, w. \tag{24}$$

The mapping $w = e^z$. Part 5

Obviously there are infinitely many points (24), since $Arg\, w$ takes infinitely many values, all differing by integral multiplies of 2π. Moreover, each of these points is actually an inverse image of w, since

$$\begin{aligned} exp[\ln|w| + i\, Arg\, w] &= e^{\ln|w|}(\cos Arg\, w + i\, \sin\, Arg\, w) \\ &= |w|(\cos Arg\, w + i\, \sin\, Arg\, w) \\ &= w \end{aligned}$$

Therefore the set of all roots of the equation $e^z = w$ ($w \neq 0$ is given by the formula

$$z = \ln\, |w| + iArgw = \ln\, |w| + i(arg\, w + 2k\pi), \tag{25}$$

where $k = 0, \pm 1, \pm 2, \ldots$.

The mapping $w = e^z$. Part 6

These points all lie on the same straight line parallel to the imaginary axis, and the distance between any two consecutive points along the line is 2π. Thus the function $w = e^z$ maps the finite z-plane onto the domain obtained from the finite w-plane by deleting the single point $w = 0$, but the mapping is not one-to-one, since every point $w \neq 0$ has an infinite number of inverse images (25). on the other hand, the mapping is conformal at every point on the finite z-plane, since the derivative

$$\begin{aligned}(e^z)' &= \frac{\partial(e^x \cos y)}{\partial x} + i\frac{\partial(e^x \sin y)}{\partial x} \\ &= e^x(\cos y + i \sin y) \\ &= e^z\end{aligned}$$

does not vanish for any value of z.

The mapping $w = e^z$. Part 7

Now suppose z traces out a straight line parallel to one of the coordinate axes. For example, consider the line

$$z = b + it, \tag{26}$$

parallel to the imaginary axis. Then the image of (26) under the mapping $w = e^z$ is the curve

$$w = e^b(\cos t + i \sin t), \tag{27}$$

i.e., w traces out a circle of radius e^b with its center at the origin. Moreover, as z describes the line (26) once in such a way that t, the ordinate of z, increases continuously from $-\infty$ to $+\infty$, w describes the circle (27) an infinite number of times in the positive (counterclockwise) direction.

The mapping $w = e^z$. Part 8

Next consider the line

$$z = t + ic \tag{28}$$

parallel to the real axis. Then the image of (28) under the mapping $w = e^z$ is the curve

$$w = e^t(\cos c + i \sin c), \tag{29}$$

i.e., w traces out a ray of slope $\tan c$ emanating from the origin. Moreover, as z describes the line (28) once in such a way that t, the abscissa of z, increases continuously from $-\infty$ to $+\infty$, w describes the ray (29) once in such a way that the distance of w from the origin increases continuously from 0 to ∞ (of course, the limits 0 and ∞ are excluded, since $|w| = e^t$).
Thus, *under the mapping $w = e^z$, a family of lines parallel to the imaginary axis is transformed into a family of concentric circles with the origin as center, and a family of lines parallel to the real axis is transformed into a family of rays emanating from the origin*.

The mapping $w = e^z$. Part 9

Now consider the domain G consisting of all points z such that

$$\phi_0 < Im\, z < \phi_1,$$

where $\phi_1 - \phi_0 = h$. Such a domain will be called *an (open) strip of width h*. Suppose $0 < h < 2\pi$, and let $\mathcal{G}$ be the image of G under the mapping $w = e^z$. It follows from the consideration just given that $\mathcal{G}$ is the interior of the angle of h radians with vertex at the origin, formed by the rays

$$Arg\, w = \phi_0 + 2k\pi, \;\; Arg\, w = \phi_1 + 2k\pi \;\; (k = 0, \pm 1, \pm 2, \ldots)$$

Moreover, the correspondence between the domains G and $\mathcal{G}$ under the mapping $w = e^z$ is one-to-one. To see this, we recall that the inverse images of a point $w \in \mathcal{G}$ are all of the form (25) lie on a line parallel to the imaginary axis, and the distance between then is an integral multiple of 2π. However, by assumption, the width h of our strip does not exceed 2π, and G can contain only one inverse image of the point w, i.e., not only is $w = f(z)$ a single-valued function on G, but its *exponential function $w = e^z$ is a one-to-one conformal mapping of an open strip of width $h \leq 2\pi$ with sides parallel to the real axis onto the interior of an angle of h radians with vertex at the origin*.

Thank you